

GENE EDITING TRAINING PIPELINE

At least five RAB early-career scientists will receive hands-on training at IITA's Gene Editing Platform at the BecA/ILRI Hub in Nairobi. The full pipeline is covered: target gene identification, guide RNA design, construct development, embryogenic cell suspension, Agrobacterium-mediated transformation, transgene-free editing, sequencing validation, and phenotypic evaluation.

BXW RESISTANCE: KAMARA

Training focuses on BXW resistance targets MusaPUB22 and MusaPUB23, already successfully knocked out in dessert banana cultivar Kamara/Sukali Ndiizi using CRISPR-Cas9. The resulting transgene-free events have been validated for mutation accuracy and absence of foreign DNA, and are now ready for field evaluation in Rwanda — led by a RAB scientist returning from her PhD at IITA in 2025.

INJAGI EDITING & FULCRUM™ SYSTEM

RAB will establish a gene editing pipeline for Injagi — a farmer-preferred cooking banana — using the FULCRUM™ system licensed by IITA from Pairwise, securing Rwanda's freedom to operate for public-sector use. RAB scientists will develop embryogenic cell suspensions for Injagi and edit susceptibility genes including Alcohol Dehydrogenase 1 (ADH1) to confer Fusarium Wilt resistance, with plans to stack both BXW and Fusarium resistance traits.

NATIONAL OWNERSHIP & LONG-TERM IMPACT

Upon training completion, RAB will have an in-country self-sustaining banana gene editing pipeline. The RAB team will lead edited line development, biosafety data generation, phenotypic evaluation under high-disease pressure, and integration into national seed systems — positioning Rwanda as a regional leader in banana biotechnology.

BioCap

BUILDING RWANDAN CAPACITY for Accelerated Development, Deregulation, and Deployment of Enhanced Crops

THE SCIENCE OF RWANDAN GROWTH

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BIOTECH BANANA

PROTECTING RWANDA'S MOST IMPORTANT FOOD SECURITY CROP

AFRICAN CROPS BY AFRICAN SCIENTISTS, FOR AFRICAN FARMERS

1. THE CHALLENGE

Banana is the backbone of Rwanda's food security, nutrition, and rural livelihoods, providing both staple food and household income for millions of smallholder farming families. However, banana production is increasingly threatened by devastating diseases, particularly Banana Bacterial Wilt (BBW) and the emerging global threat of Fusarium Wilt Tropical Race 4 (TR4). Banana Bacterial Wilt can destroy entire plantations if not controlled, while TR4 has devastated banana production in several regions of the world and poses a serious risk to East Africa.

Current disease management depends largely on field sanitation, destruction of infected plants, and continuous surveillance. These approaches are labor-intensive, costly, and often difficult for smallholder farmers to sustain. The frequent loss of banana plants reduces household food availability, farmer income, and national agricultural productivity while increasing the cost of maintaining healthy plantations.

2. THE BIOTECHNOLOGY SOLUTION

Through the Rwandan Agricultural Biotech Capacity Building programme, the investment strengthens Rwanda's ability to develop the next generation of biotechnology crops locally. For banana, this means building RAB capacity to generate, evaluate, and steward **farmer-preferred banana varieties with resistance to Banana Bacterial Wilt and Fusarium Wilt Tropical Race 4,** supported by national laboratory, greenhouse, and regulatory systems.

The investment document emphasizes self-reliance: Rwanda will not only receive biotech products, but also build the scientific capacity to create, test, document, and deploy them responsibly. Biotechnology can introduce targeted disease-resistance traits while preserving the agronomic and culinary qualities Rwandan farmers and consumers already value.

Expected benefits include:

- Reduced disease losses
- Improved household food security
- Increased farmer incomes
- Protection of local banana varieties
- Greater resilience against future disease outbreaks

3. PARTNERSHIP FOR INNOVATION

The Banana Biotechnology Programme is implemented through a strong international partnership:



4. BUILDING RWANDA'S FUTURE

Centre of Excellence in Crop Biotechnology: Banana Platform

The investment will establish and strengthen a Centre of Excellence for Crop Biotechnology at RAB, able to conduct genetic transformation, genome editing, advanced molecular diagnostics, contained evaluation, and regulatory-quality data generation. The banana platform contributes to this national capacity by developing safe, traceable workflows for banana improvement and future disease-resistance innovation made in Rwanda.

For banana, the centre should be presented as a controlled, professional research environment with biosafety-compliant laboratory workflows, contained plant transformation and tissue culture rooms, molecular diagnostics capacity, screenhouse/greenhouse evaluation space, and standard operating procedures for safe handling, documentation, traceability, and responsible technology transfer to RAB scientists and technicians.



